

(2026) find similar adoption patterns for generative AI across the US, UK, Germany, and Australia.⁵ While firm size is unrelated to encouraged-use policies—perhaps due to the low fixed costs of off-the-shelf chatbot solutions—larger firms are more likely to purchase enterprise versions.

Overall, however, chatbot initiatives are largely unrelated to observable firm characteristics, which account for just 1.3% of the within-occupation variation in encouragement—far less than the 10.6% explained by our 11 occupation categories alone. In other words, many otherwise comparable firms have adopted markedly different strategies toward AI chatbots, even for similar workers within the same occupation.

2.1.1 Employer Initiatives and Worker Adoption

We next examine how workers’ adoption of and benefits from AI chatbots vary with the employer-led chatbot initiatives. Specifically, we compare workers in the same occupations with similar characteristics and estimate how their adoption behaviors and reported benefits vary with employer initiatives for AI chatbots:

$$Y_i = \gamma'X_i + \beta' \text{EmployerInitiatives}_i + \varepsilon_i, \quad (1)$$

where Y_i is a chatbot-related outcome for worker i , such as use of or reported benefits from AI chatbots; X_i includes survey occupation fixed effects and worker pre-determined characteristics, including age, gender, and potential experience;⁶ and $\text{EmployerInitiatives}_i$ is a vector of employer-driven initiatives, including encouraging employees to use AI chatbots, investing in enterprise chatbot solutions, providing employee training—and combinations thereof. We estimate Equation (1) by OLS and cluster the standard errors at the workplace-occupation level.

The specification in Equation (1) is motivated by a large literature on complementarities between new technologies and organizational investments (Bloom, Sadun and Reenen, 2012;

⁵Younger and more productive firms are commonly found to adopt new technologies faster; see, e.g., Acemoglu et al. (2022), who document adoption patterns for a range of advanced technologies in 2019, including AI and robotics.

⁶All worker characteristics are measured in 2022, before AI chatbots became available.

Brynjolfsson and Hitt, 2000), which emphasizes that the returns to technology adoption may depend critically on firms’ broader organizational investments and management practices. Moreover, because early adoption of AI chatbots was largely worker-driven, this analysis sheds light on how adoption and work-related benefits differ in workplaces where employers also invest in the technology.

Appendix A.1 presents a Roy-style model of chatbot adoption to help interpret the estimates in Equation (1). The model shows that β for user-reported benefits combines an *individual*-level effect of the initiative on each user’s gains with a *selection* effect arising from shifts in which workers opt into use. When workers adopt based on their individual gains, selection will tend to be negative: initiatives that expand adoption draw in workers who benefit less from the technology. We do not attempt to separate these two effects, as we view the bundled impact of employer initiatives as informative in itself.

While we interpret Equation (1) as descriptive, our rich firm- and worker-level data allow us to assess the robustness of these patterns to a range of potential confounders. Appendix F.5 presents these robustness checks. First, we show that all results hold when controlling for firm characteristics, ensuring that differences across employer initiatives are not driven by variation in firm age, size, or productivity. Second, we show that the results are robust to controlling for workers’ detailed task mixes within occupations, demonstrating that the effects of employer initiatives are not driven by differences in the types of tasks more amenable to AI chatbot use.⁷ In addition, the panel analyses in Section 3 confirm that adopters of different initiatives followed similar labor market trajectories prior to the arrival of AI chatbots. Together, these robustness checks help rule out confounding factors and support the validity of our estimated relationships.

Since our data are collected from workers rather than senior leadership, our estimates reflect employer initiatives *as perceived by workers*. Since employer policies may not be perfectly communicated, worker self-reports may not fully align with employer intentions.

⁷We do not include task controls in our main specification, as task mix may itself be endogenous to chatbot adoption. Indeed, Section 2.3 shows that AI chatbots have created new job tasks for many workers.

Nonetheless, coworker reports are tightly correlated within workplaces, and Appendix F.6 shows that our findings are robust to using coworker reports to measure employer initiatives, supporting their interpretation as workplace-wide policies.

2.2 Worker Adoption and Reported Benefits

Figure 2 shows how AI chatbot adoption varies with employer initiatives for a typical worker. The top estimate (*No Initiative*) highlights the bottom-up nature of chatbot diffusion: even among workers whose employers neither encourage chatbot use, provide an enterprise chatbot, nor offer training, about 41% have used AI chatbots at work at least once, and 7% use them daily.

Despite this high baseline, employer initiatives are associated with substantially higher adoption. Implemented individually, *Encouraged use*, *Enterprise chatbots*, and *Training* each raise extensive-margin take-up rates to 76%, 57%, and 69%, respectively. Encouraged-use policies are associated with especially high intensive usage, with daily use reaching 18%.

Table 1 shows that encouraged-use policies are not only associated with higher chatbot usage (Column 1-2), they are also consistently associated with users reporting greater time savings, improved quality, and enhanced creativity (Columns 3-6).⁸

Finally, as the bottom row of Figure 2 shows, adoption peaks when all three—encouragement, enterprise chatbots, and training—are in place: 93% of workers in such settings report having used AI chatbots at work, and 28% use them daily.

Discussion. The finding that adoption and reported benefits are greatest when AI chatbots are combined with employer initiatives aligns with a large literature on complementarities between new technologies and organizational investments, including people management practices (Bloom, Sadun and Reenen, 2012; Brynjolfsson and Hitt, 2000).

⁸By contrast, training and enterprise tools—when implemented without encouragement—are associated with smaller reported gains. This pattern is consistent with these initiatives being aimed at mitigating misuse rather than enhancing productivity. Supporting this interpretation, Section 2.3 shows that *AI Ethics & Compliance* tasks are especially prevalent among workers who have received employer-provided training.

Still, the relative importance of encouraged-use policies may seem surprising, as such policies might appear inexpensive in monetary terms. But the low capital costs of chatbot solutions (Appendix E.2 details adoption by chatbot products) heighten the importance of complementary organizational practices, such as explicitly encouraging use. For example, Shopify has publicly reframed generative AI from something employees may use into a baseline expectation; Chicago Public Schools explicitly encourages faculty to engage with GenAI tools; and Denmark’s Agency for Digital Government similarly emphasizes that adoption requires clear usage guidelines and supportive organizational frameworks even when publicly available tools are used (Chicago Public Schools, 2024; Digitaliseringsstyrelsen, 2024; Lütke, 2025). Consistent with this interpretation, Dillon et al. (2025) show that firm effects are a key determinant of workers’ uptake of a randomized offer of Microsoft Copilot.

2.3 New Workloads and Task Creation

AI chatbots not only affect existing tasks—they transform the nature of work itself. Figure 3 shows how the emergence of new workloads from AI chatbots varies across employer initiatives. These new workloads may stem from individual workers taking on more work, from shifts in task allocation within teams, or from new demands generated by the tools themselves, such as oversight and integration.

The *No Initiative* estimates reveal that even in the absence of encouraged-use policies, enterprise solutions, or training, AI chatbots have generated new workloads for 12% of users (Panel (a)). Of these, about 5 percentage points report doing more of the same tasks, and 8 percentage points have taken on entirely new tasks.⁹

Notably, the share of users performing new tasks in Figure 3 increases to roughly 18% in workplaces with active chatbot initiatives. Training, in particular, is strongly associated with new job responsibilities, suggesting that these programs may be designed to help

⁹Consistent with this reorganization of work, most chatbot users (85%) report reallocating time savings from AI chatbots to other job tasks (Table E.3). By contrast, fewer than 10% report taking additional breaks or leisure, and about 30% spend more time on the same tasks they initially saved time on.